ENVODEX CONSULTING ENGINEERS

Building Envelope & Facade Structural Engineering Consultancy — *Company Profile 2026*

Overview & Philosophy

Envodex is an independent, specialized engineering consultancy delivering high-precision structural analysis, Finite Element Analysis (FEA), connection engineering, and building physics for complex architectural envelopes worldwide.

Core Computational Software Suite

- **Strand7 (Straus7):** Advanced non-linear structural modeling, plate stress analysis, dynamic response, and detailed finite element brackets.
- **STAAD.Pro:** Space frame analysis, secondary steel structures, primary mullion/transom grid structural checks.
- **SJ Mepla:** Laminated, monolithic, and insulated glass stress, deflection, and post-breakage verification under wind and thermal loads.
- **LBNL THERM:** 2D thermal bridging, U-value quantification, condensation risk, and thermal performance evaluations.

International Design Standards Compliance

- **CWCT Standards:** Center for Window and Cladding Technology guidelines for weathertightness, impact testing, and system safety.
- **Eurocodes:** EN 1991 (Actions/Wind loads), EN 1993 (Steel structures), EN 1999 (Aluminum structural design), prEN 19100 (Glass structures).
- **ASTM Standards:** ASTM E1300 (Glass load resistance), ASTM E2751 (Walkable glass floors), and related facade structural criteria.

Landmark Project Portfolio

- **Chic Tower (Dubai):** Structural engineering and custom curtain wall profile verification for ultra-luxury residential skyscraper.
- **Masdar City (Abu Dhabi):** Sustainable building envelope engineering, high-performance shading louvers, and extreme thermal resilience.
- **One North Quay (London, UK):** High-rise science and commercial tower envelope structural package and dynamic wind load analysis.
- **220 Blackfriars Road (London, UK):** Detailed EWS bracket connection calculations, mullion splice analysis, and fabrication design verification.
- **KAPSARC University (Riyadh, KSA):** Complex geometric envelope coordination, secondary framing, and high-performance paneling systems.

Key Client Partnerships

Collaborating with international industry leaders including **Permasteelisa Group**, **Future Glazing**, **Realiteza**, and **Global Facade Solutions (GFS)**.

Contact Information

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